Graph Neural Networks for Molecular Property Prediction

Abstract

We present a deep learning approach using Graph Convolutional Networks (GCN) to predict molecular toxicity with high a

Introduction

Predicting molecular properties early in the drug discovery pipeline reduces costs and accelerates development. Traditior

Methodology

Molecules are represented as undirected graphs G=(V,E) where nodes V represent atoms and edges E represent chemic

Results

Our model achieves AUC-ROC of 0.94 on Tox21 and 0.91 on BBBP. Ablation studies confirm attention-based readout co

Conclusion

Graph neural networks are powerful tools for molecular property prediction. The proposed architecture generalizes across

References

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